producing a loud, high pitched note.

2 a What is an echo?

b Explain how echoes are used by ships to find the depth of the ocean beneath them.

c A ship hears the echo from a sound wave 4 s after it has been emitted. If the speed of sound in water is 1500 m/s, calculate the depth of the water beneath the ship.

3 a What is meant by the phrase 'a person's audible range is 20 Hz to 20 000 Hz'?

b Explain why the vibrating strings of a violin produce sounds with a higher frequency than those produced by the strings of a cello.

c Calculate the wavelength of sound waves whose frequency is 68 000 Hz. Assume that the waves are travelling through air at a speed of 340 m/s.

4 a An oscilloscope shows a wave which has a time period of 0.01 s. What is the frequency of this wave?

b If the speed of this wave is 340 m/s, calculate its wavelength.

5 a Sound waves are emitted from a source that is vibrating with a large amplitude and from a source that is vibrating with a small amplitude. Explain, using diagrams, the difference between the two sets of sound waves.

b Draw two diagrams to show how these waves would appear on an oscilloscope.

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